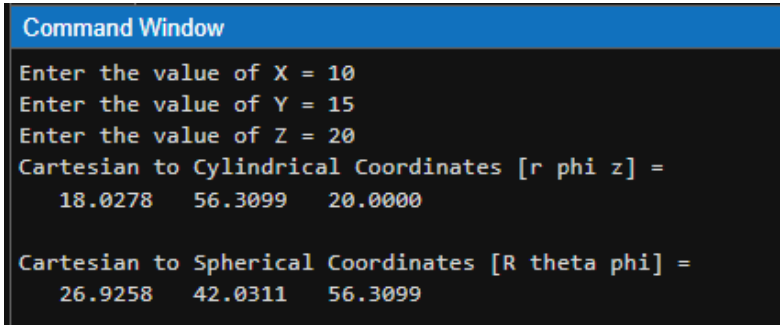


Experiment 1.1

Transformation between Cartesian, Cylindrical and spherical coordinate system

Code File Name
Exp1_1_Cart2CylSph (1).m
Code
<pre>clc; clear; close all; x = input('Enter the value of X = '); y = input('Enter the value of Y = '); z = input('Enter the value of Z = '); % Cartesian to Cylindrical r = sqrt(x^2 + y^2); phi = atan2d(y, x); disp('Cartesian to Cylindrical Coordinates [r phi z] ='); disp([r phi z]); % Cartesian to Spherical R = sqrt(x^2 + y^2 + z^2); theta = acosd(z / R); phi = atan2d(y, x); disp('Cartesian to Spherical Coordinates [R theta phi] ='); disp([R theta phi]);</pre>
Command Prompt Output
<pre>Enter the value of X = 10 Enter the value of Y = 15 Enter the value of Z = 20 Cartesian to Cylindrical Coordinates [r phi z] = 18.0278 56.3099 20.0000 Cartesian to Spherical Coordinates [R theta phi] = 26.9258 42.0311 56.3099</pre>
Attached Command Window Screenshot
 <pre>Command Window Enter the value of X = 10 Enter the value of Y = 15 Enter the value of Z = 20 Cartesian to Cylindrical Coordinates [r phi z] = 18.0278 56.3099 20.0000 Cartesian to Spherical Coordinates [R theta phi] = 26.9258 42.0311 56.3099</pre>